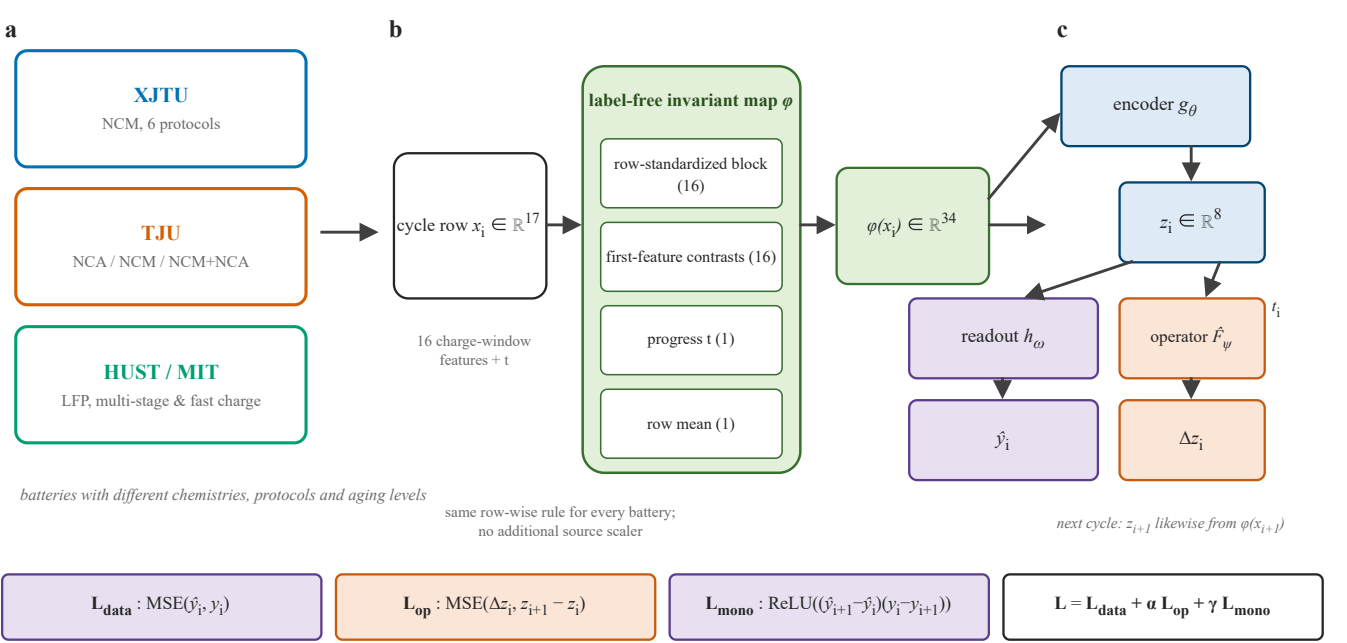

a
XJTU
NCM, 6 protocols
TJU
NCA / NCM / NCM+NCA
HUST / MIT
LFP, multi-stage & fast charge
batteries with different chemistries, protocols and aging levels
b
cycle row $x_i \in \mathbb{R}^{17}$
16 charge-window features + t
label-free invariant map φ
row-standardized block (16)
first-feature contrasts (16)
progress t (1)
row mean (1)
same row-wise rule for every battery; no additional source scaler
$\varphi(x_i) \in \mathbb{R}^{34}$
c
encoder g_θ
$z_i \in \mathbb{R}^8$
t_i
readout h_ω
operator $\hat{F}_\psi$
$\hat{y}_i$
Δz_i
next cycle: z_{i+1} likewise from $\varphi(x_{i+1})$
$\mathbf{L_{data}}$: MSE($\hat{y}_i$, y_i)
$\mathbf{L_{op}}$: MSE(Δz_i, $z_{i+1} - z_i$)
$\mathbf{L_{mono}}$: ReLU($(\hat{y}_{i+1} - \hat{y}_i)(y_i - y_{i+1})$)
$\mathbf{L = L_{data} + \alpha\, L_{op} + \gamma\, L_{mono}}$